

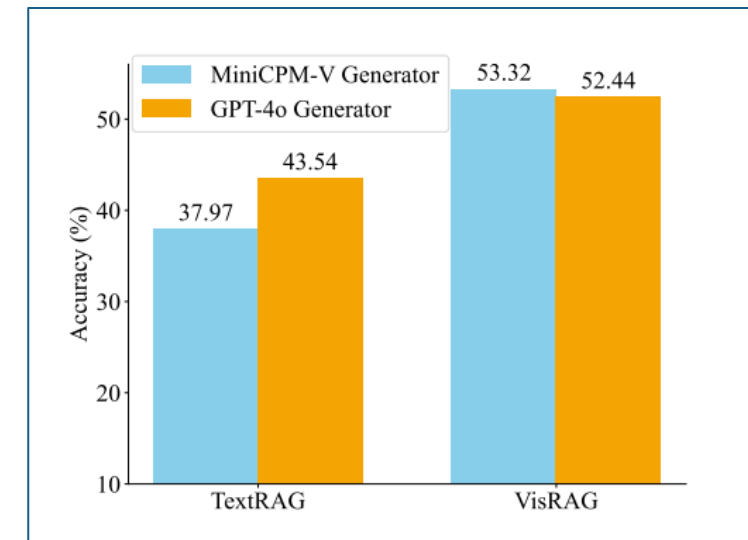
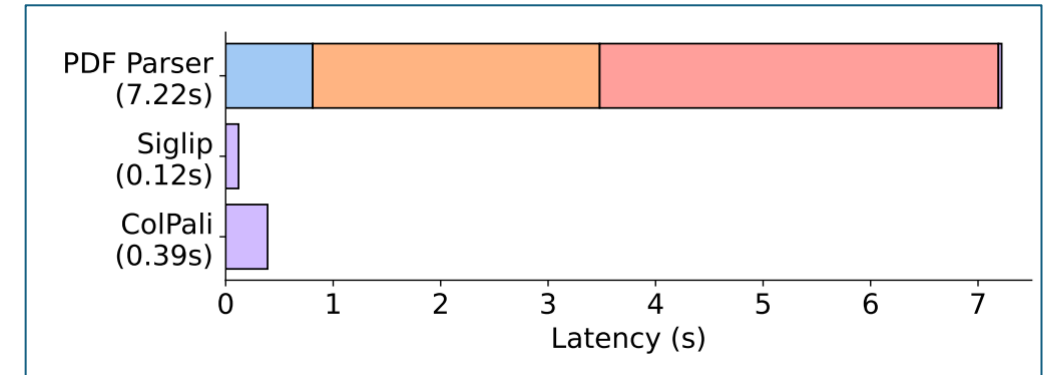
One Pic is All it Takes:

Poisoning Visual Document Retrieval Augmented Generation with a Single Image

Ezzeldin Shereen, Dan Ristea, Burak Hasırcioğlu, Shae McFadden,
Vasilios Mavroudis, Chris Hicks

Visual Document RAG

- PDF documents are widely-used sources of knowledge for RAG
 - Not only text
- Traditional document parsing through **OCR+Captioning pipelines are inefficient**
- Recent works proposed **direct retrieval using VLMs** (i.e., VDR)^[1]
- VD-RAG is **20-40% more accurate** than text-based RAG^[2]

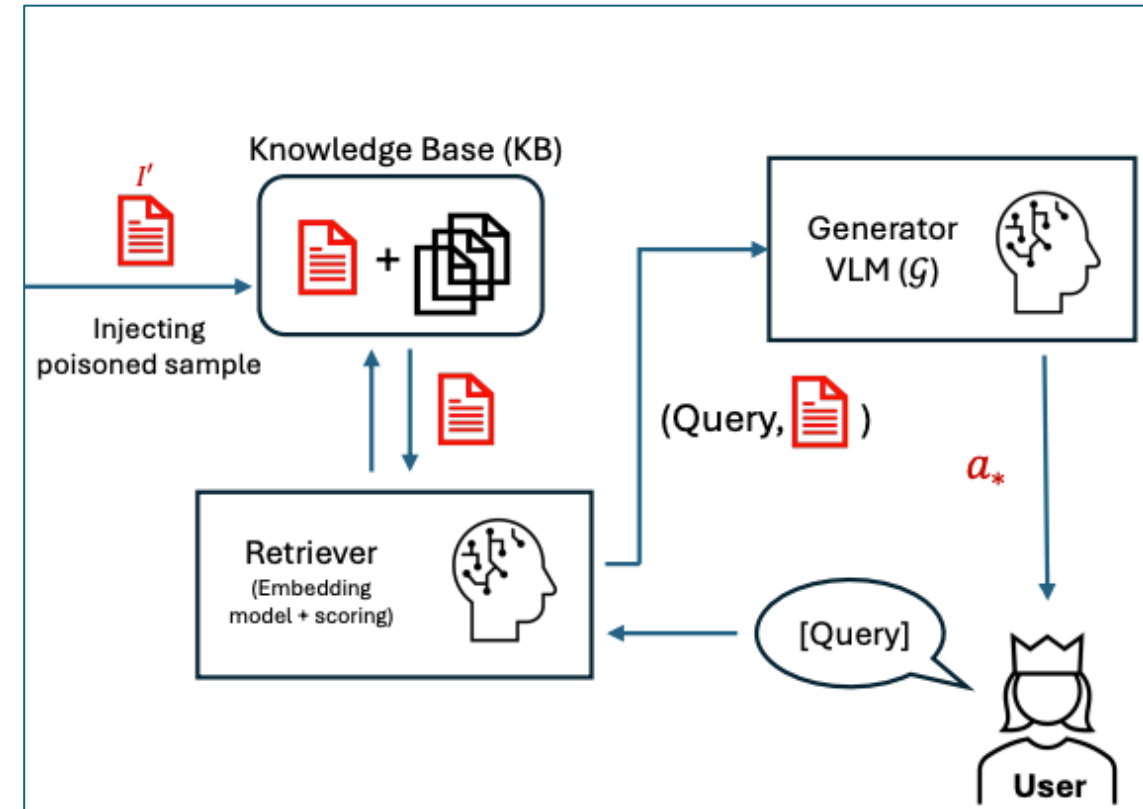


[1] Faysse, Manuel, et al. "Colpali: Efficient document retrieval with vision language models." arXiv preprint arXiv:2407.01449 (2024).

[2] Yu, Shi, et al. "Visrag: Vision-based retrieval-augmented generation on multi-modality documents." arXiv preprint arXiv:2410.10594 (2024).²

Security of VD-RAG

- Adversaries can **inject malicious documents** into knowledge base
 - Insider vs. outsider threats
- Previous work
 - RAG poisoning attacks in text domain
 - *PoisonedRAG*^[1], *BadRAG*^[2]
 - Attacks in multi-modal domain (text-image pairs)
 - *MM-PoisonRAG*^[3] (Feb 2025), *Poisoned-MRAG*^[4] (Mar 2025)



[1] Zou, Wei, et al. "PoisonedRAG: Knowledge Corruption Attacks to Retrieval-Augmented Generation of Large Language Models." arXiv preprint arXiv:2402.07867 (2024)

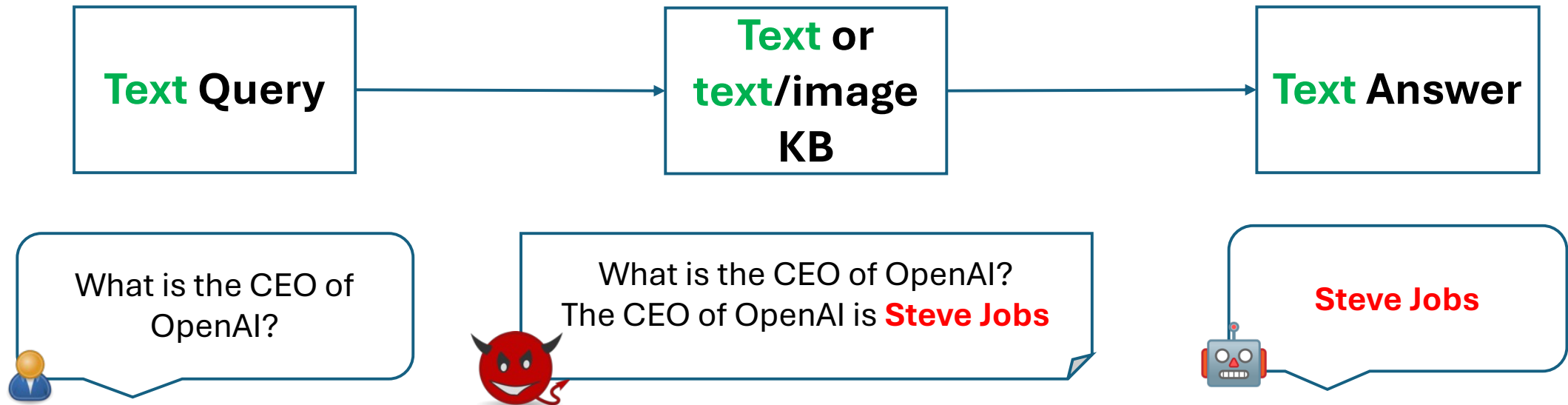
[2] Xue, Jiaqi, et al. "Badrag: Identifying vulnerabilities in retrieval augmented generation of large language models." arXiv preprint arXiv:2406.00083 (2024).

[3] Ha, Hyeonjeong, et al. "MM-PoisonRAG: Disrupting Multimodal RAG with Local and Global Poisoning Attacks." arXiv preprint arXiv:2502.17832 (2025).

[4] Liu, Yinuo, et al. "Poisoned-MRAG: Knowledge Poisoning Attacks to Multimodal Retrieval Augmented Generation." arXiv preprint arXiv:2503.06254 (2025). 3

What's different about VD-RAG?

- Existing attacks **all utilize text** as part of the attack
 - Allows for trivial attacks



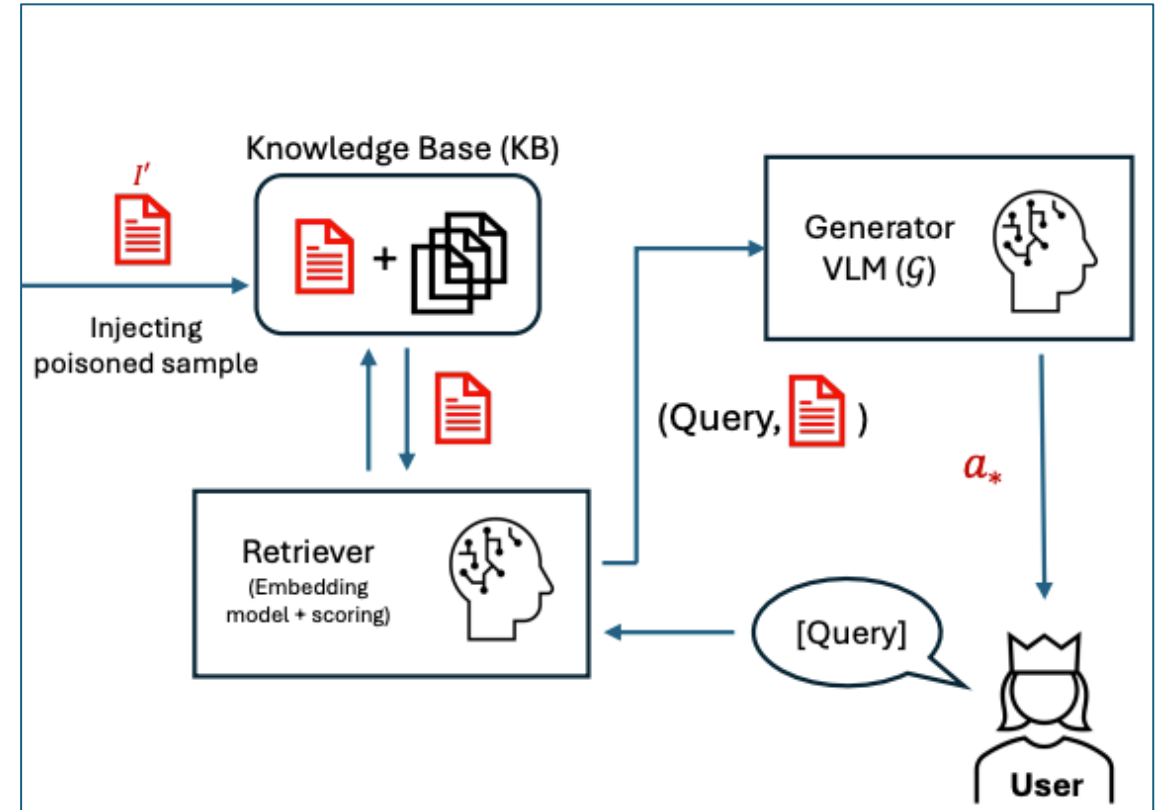
Proposed Attack

- Adversary injects **a single image** to the knowledge base
- Attack goal
 - **Retrieval Condition:** The injected image is retrieved as relevant for all user queries
 - **Generation Condition:** When retrieved, the image influences the VLM generated outputs
- Image optimized using the gradient-based adversarial approach

$$\mathcal{L}_{RAG} = \lambda_R \mathcal{L}_R + \lambda_G \mathcal{L}_G$$

Distance between image embedding and all query embeddings

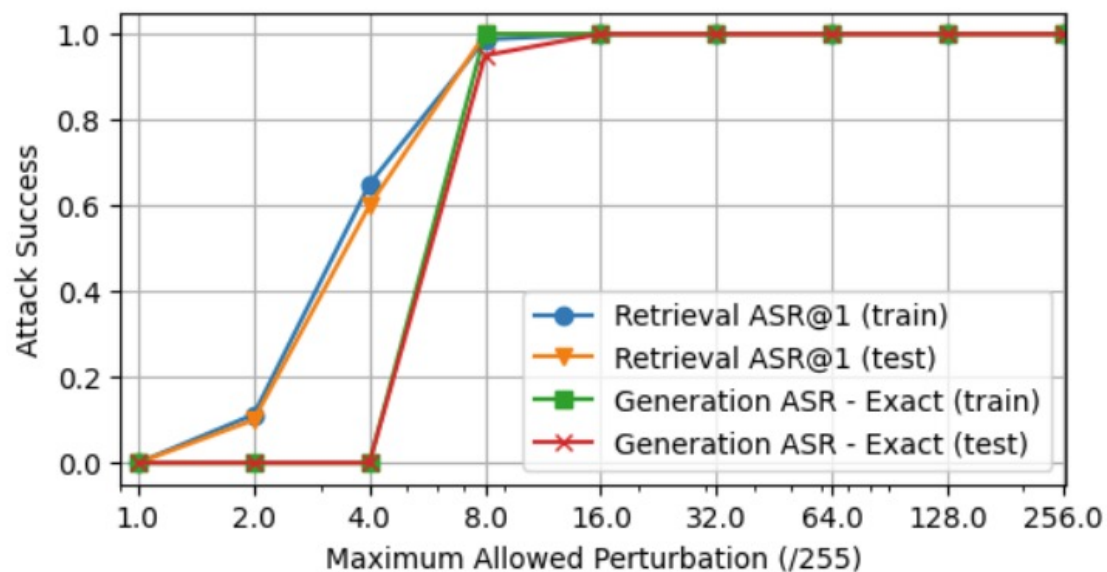
CE loss for target VLM answer



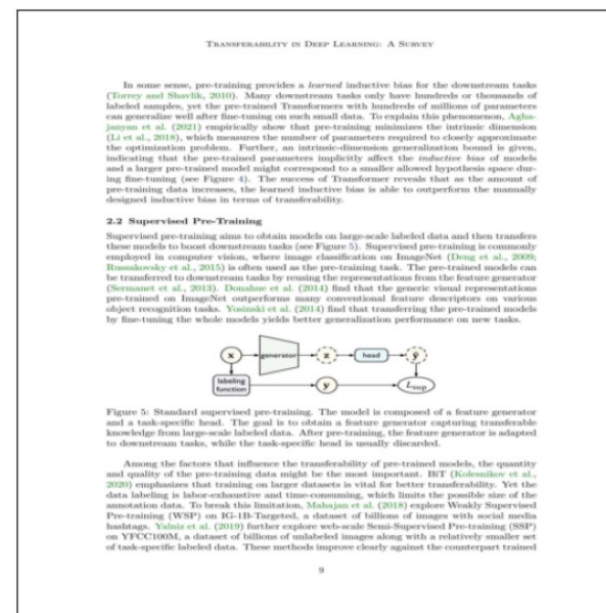
Experiments

Datasets	<i>ViDoRe-v1-AI</i>	100 questions and 1000 images from IT and AI related documents (1:1)
	<i>ViDoRe-v2-ESG</i>	52 queries and 1538 images from restaurant related documents (1:many)
Embedders	<i>CLIP-LARGE</i>	Seminal 0.4B model from OpenAI
	<i>Jina-CLIP-v2</i>	0.9B recent model trained to bridge the modality gap
	<i>ColPali</i>	3B SoTA model for VDR
VLMs	<i>SmolVLM-Instruct</i>	2.2B lightweight model by HuggingFace
	<i>Qwen2.5-VL-3B</i>	3.75B SoTA open-weights VLM

Attack Stealthiness



- Trained against CLIP-L and Smol-VLM
- Attack is successful starting from a perturbation of (8/255)
- Perturbation barely visible at (8/255)



Results – ViDoRe-V1-AI

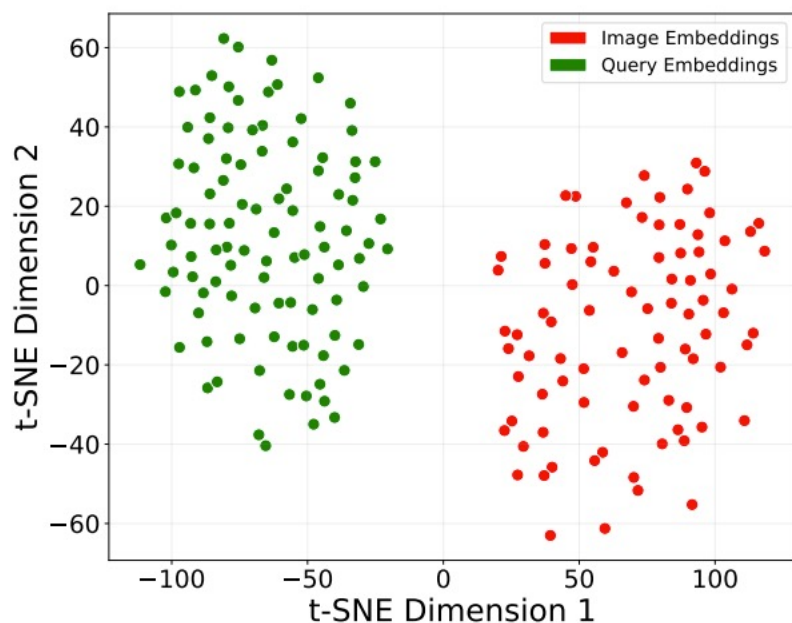
Embedder	VLM	ASR-R	ASR-G*
CLIP-L	SmolVLM	1	1
CLIP-L	Qwen2.5-3B	1	1
Jina-CLIP	SmolVLM	0.75	0.9
Jina-CLIP	Qwen2.5-3B	0.75	0.95
ColPali	SmolVLM	0	0.9
ColPali	Qwen2.5-3B	0	1

- Non-SoTA embedding models are vulnerable
- Both VLMs are vulnerable
- No black-box transferability

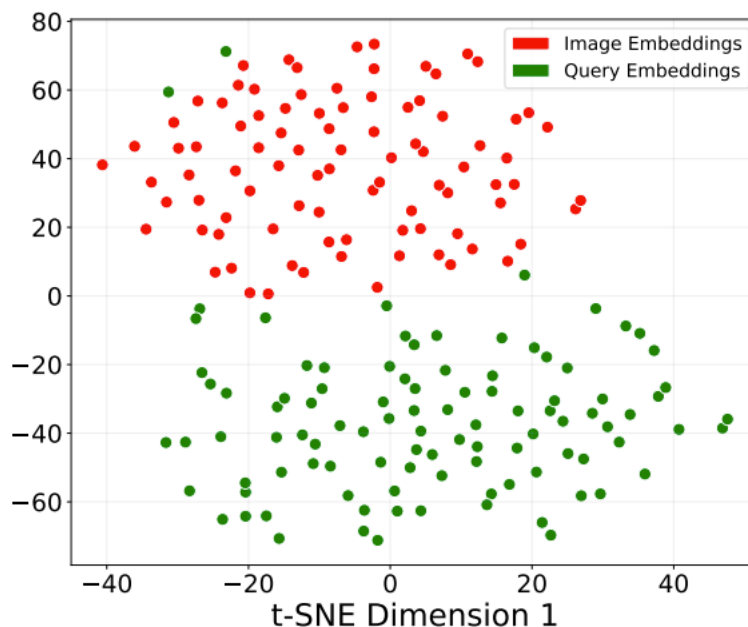
* Results for generation assume successful malicious retrieval

Explanation – The Modality Gap^[1]

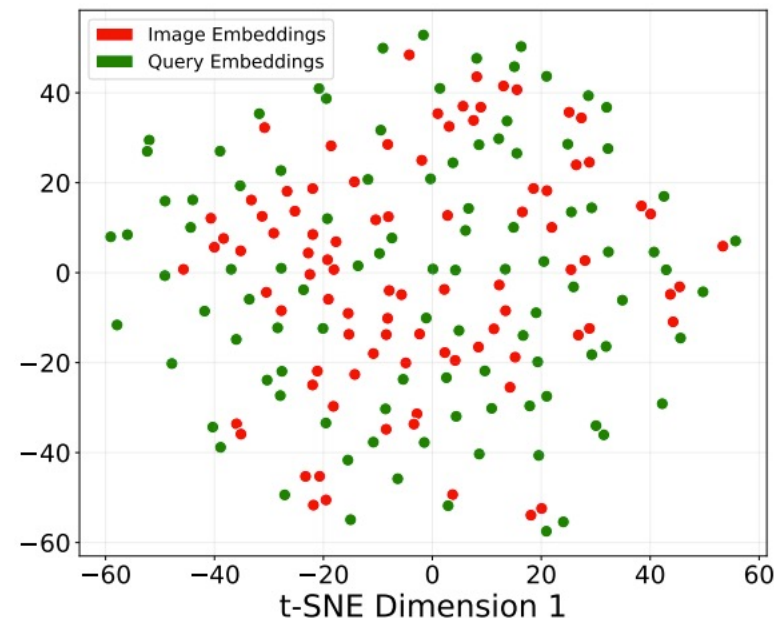
CLIP-L



Jina-CLIP



ColPali



[1] Liang, Victor Weixin, et al. "Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning." Advances in Neural Information Processing Systems 35 (2022): 17612-17625.

One Pic is All it Takes:

Poisoning Visual Document Retrieval Augmented Generation with a Single Image

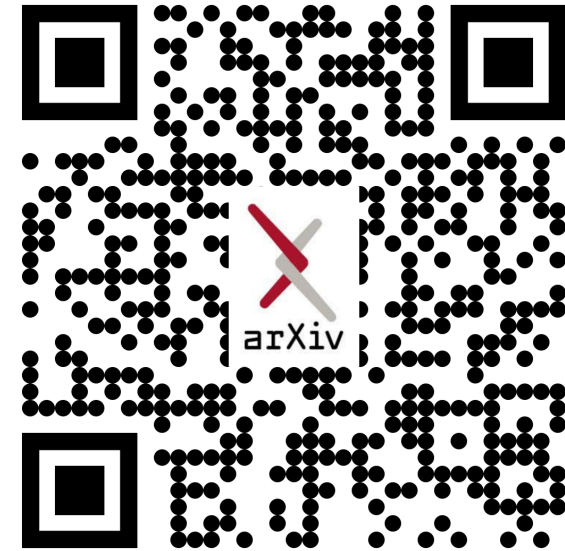
- **Takeaways**

1. VD-RAG is promising, but ..
2. One pic is all it takes!
3. Mind the modality gap!

- **Future Work**

1. Defenses, and adaptive attacks
2. Targeted attacks (stealthier)
3. Larger and more SoTA models

Extended
version



Code

